

Mohamed Yassin

DevOps Engineer

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Summary

B.Sc. in Computer Science | Military Status: Exempted

Results-driven Software Engineer with extensive experience across the full Software Development Life Cycle (SDLC), specializing in backend architecture, robust API design principles, and scalable distributed systems using Node.js and TypeScript. Proficient in database management systems. Experienced working in Agile environments (Scrum, Kanban) collaborating with cross-functional teams to design, code, test, debug, and deliver high-availability cloud solutions on AWS and GCP.

Work Experience

Software Engineer Intern – Egyptian Space Agency (EgSA)

Aug 2025 – Nov 2025

- Designed, coded, and deployed high-performance backend microservices and RESTful APIs adhering to API design best practices, processing satellite imagery to support large-scale land-use analytics.
- Collaborated closely with cross-functional teams in an Agile (Scrum/Kanban) workflow, participating in sprint planning, code reviews, and user requirements analysis from concept to delivery.
- Built automated CI/CD pipelines (GitHub Actions) with comprehensive testing and debugging gates (unit/integration tests, linting), significantly reducing release defects.
- Managed containerized deployments using Docker and Kubernetes (GKE) on cloud infrastructure.

Projects

PeerLink – Distributed File Sharing Platform

[repo](#) [🔗](#)

- Architected a fault-tolerant distributed system utilizing Node.js, TypeScript, PostgreSQL (SQL), and Redis (NoSQL) with high-availability clustering, automated failover, and Redis leader election.
- Implemented RESTful APIs applying strict API design principles and rate limiting.

HoneyKube – AI-Powered Kubernetes Honeypot

[repo](#) [🔗](#)

- Developed an LLM-driven honeypot platform using Python, Kubernetes, and Docker to aggregate and analyze threat telemetry; designed robust container networking and automated debugging pipelines.

GUIShell – High-Performance Remote Desktop Engine

[repo](#) [🔗](#)

- Built a low-latency remote desktop engine using Electron, WebRTC, and Node.js featuring custom MJPEG-over-DataChannel streaming protocols, backed by an optimized SQLite signaling server.

Key Achievements

- [TurboNodeIO](#) [🔗](#): Engineered a native C++ acceleration layer for Node.js, achieving up to 317× faster I/O throughput and 71× faster SIMD checksum processing, demonstrating advanced systems optimization and debugging.

Technical Skills

Languages: Node.js, TypeScript, JavaScript, Python, C++, SQL, Bash

Backend & API Design: RESTful APIs, API Design Principles & Best Practices, Microservices Architecture, Testing & Debugging (Jest, Supertest), System Design, Scalability

Databases & Search: SQL (PostgreSQL, SQLite), NoSQL (Redis, MongoDB), Elasticsearch (Search & Analytics), Prisma ORM

Cloud & DevOps: AWS (S3, EC2), GCP, Docker, Kubernetes (GKE), CI/CD (GitHub Actions, Jenkins), Git, Linux, Nginx

Methodologies & Practices: Agile Methodologies (Scrum, Kanban), Full SDLC, Code Reviews, Observability (Prometheus, Grafana)

Education

Modern Academy – B.Sc. in Computer Science

Sept 2022 – May 2026

- Cumulative Grade:** Excellent (A-) with Second-Class Honors (GPA: 3.29 | 89.76%)
- Graduation Project:** Excellent (A+) – Autonomous checkout system leveraging zero-shot computer vision inference without custom datasets; designed scalable backend ingestion pipelines.

Additional Skills & Activities

Soft Skills: Cross-Functional Collaboration, Problem-Solving, Technical Communication, Continuous Learning, Team Leadership

Activities & Contests: Competitive Programming (ECPC Participant, Codeforces, LeetCode), ICMTC CTF Participant

Languages: Arabic (Native), English (Fluent)